Quantifying the Cost of Delayed Energization (CoDE) for AI Data Centers

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***Abstract*—Grid interconnection has become a binding constraint on artificial intelligence (AI) data center development, yet existing planning frameworks do not price service that arrives later than planned. This paper develops the Cost of Delayed Energization (CoDE), a present-value framework that maps delayed information technology (IT) capacity to grid MW and MVA requirements and prices the delay separately for developers, ratepayers, and society. Two decision metrics, delay cost intensity and the break-even acceleration payment, make results comparable across project sizes, delays, and mitigation options. For a 300 MW-IT project delayed 36 months, private CoDE is 714 million USD, or 66.1 thousand USD per MW-IT-month, and committed-capital carrying cost keeps it above 1.0 billion USD at 60 months even as realization probability decays. A one-year natural-gas bridge over the same delay avoids 257 million USD of private cost while adding 25–68 million USD of expected climate damage, and the delay itself adds 25 million USD of ratepayer exposure. Because these third-party effects are smaller than the private values they are set against by factors of about four to thirty, the operative planning question is who should fund acceleration and how pathway externalities are compensated.**

***Index Terms*—AI data centers, delayed energization, grid interconnection, large-load planning, tariff design.**

I. Introduction

Data centers are now among the largest and fastest-growing electricity loads. Efficiency gains historically decoupled rising digital service demand from electricity use, but future demand remains sensitive to technology and deployment trajectories [1], [2]. A recent review of artificial intelligence (AI) data centers extends this concern to grid planning, operations, and reliability [3]. U.S. data centers consumed about 4.4% of national electricity in 2023 and could reach 6.7–12% by 2028 [4]; a separate outlook projects 9–17% of national electricity and 56–132 GW of nominal information technology (IT) capacity by 2030 [5]. Reliability assessments now treat such loads as material drivers of demand growth and planning uncertainty [6].

Industry market reports show the same constraint from the project side. CBRE finds record wholesale pricing and limited powered (grid-connected) capacity [7], and JLL identifies power constraints and multi-year waits for grid connection as central site selection issues [8]. Energization timing therefore determines whether committed computing capacity earns revenue or sits idle, and whether network spending on its behalf is recovered.

Delayed grid energization is a planning problem in its own right. A large load may be commercially committed before it receives first usable grid service, and industry terms such as *speed to power* name the schedule pressure without saying how delay costs should be valued or allocated. Interruption cost models value outages for already-connected customers [9], [10]; adequacy and interconnection studies address network and supply-side capacity interactions [11]; queue studies quantify delays for generation assets [12]. A delayed data center differs from all of these because its economic consequences begin before operation.

Market reports give project-level proxies for the scarcity of powered capacity, lease value, and capital cost [7], [8], and flexibility papers show that computing load can respond to prices, demand response signals, and dispatchability opportunities [13]–[16]. Large-load tariff and governance sources identify cost shifting, stranded asset risk, speculative requests, and protection mechanisms [17], [18], and public tariffs and dashboards show mechanisms such as collateral, minimum bills, and separate large-load rates [19]–[21]. None of this work prices a delay that begins before first service.

The gap is practical becasue pending large-load tariff dockets must already decide who bears network investment risk. Minimum bills, collateral, exit fees, and contributions in aid of construction reduce residual ratepayer exposure but do not measure the developer's private value of earlier service, while a high private willingness to pay establishes no public benefit if residual network costs or externalities remain outside the payment. A useful framework must therefore keep private value, ratepayer exposure, and social effects distinct, and state the rule under which they may be recombined.

This paper develops the Cost of Delayed Energization (CoDE), a framework that prices the interval a planned large load spends awaiting first usable grid service, separately for developers, for ratepayers, and for society. Table I summarizes its focus relative to existing analysis frameworks. The private term is intentionally conventional, namely discounted foregone cash flows plus carrying cost on committed capital [22]. The contribution lies in the ratepayer and pathway-externality terms that project finance omits, and in metrics that make all three comparable in a tariff proceeding. The contributions are

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TABLE I
CoDE Relative to Existing Analysis Frameworks

Framework	Main object	CoDE's focus
Interruption cost	An energized customer loses service.	Values delay before service, not restoration.
Generation queue	A supply asset waits to interconnect.	Values a delayed load and its commercial exposure.
Construction delay	A project schedule slips.	Isolates energization as the binding constraint.
Large-load tariff	Upgrade and stranded risk are allocated.	Separates private, ratepayer, and social delay value.

as follows.

- Define delayed grid energization as a pre-service planning problem and introduce the CoDE perspectives, delay cost intensity (DCI), and break-even acceleration payment (BEAP).
- Separate project realization, substitution, committed-capital exposure, residual ratepayer risk, and pathway externalities, and state an explicit net-benefit test for recombining them.
- Apply the workflow to AI data center benchmarks, including parameter sweeps, delay–realization dependence, bridge power, staged service, and tariff risk.

The rest of this paper is organized as follows. Section II presents the CoDE framework. Section III presents the case study results, and Section IV concludes the paper.

II. CoDE Framework

A. Framework Overview and Perspectives

Delayed grid energization is the interval between a planned service date and first usable grid service; hereafter, the delay. Unlike an outage or a queue delay, it concerns a load request that may already be tied to commercial commitments, upgrade planning, or tariff protections before load is served. As shown in Fig. 1, CoDE maps delayed IT capacity and delay duration to facility-level grid MW and interconnection MVA, assigns the resulting costs to private, ratepayer, and social perspectives, and normalizes each into per-MW, per-month metrics.

Let P_I denote delayed IT capacity in MW-IT, representing the server and networking load, and let τ be the delay in months. The facility-level grid demand is $P_G = \mathrm{PUE}\, P_I$, where power usage effectiveness (PUE) expands IT load to total data center load, and interconnection apparent power is $S = P_G/\mathrm{pf}$, where pf is the power factor. CoDE is defined as a vector of present-value quantities by perspective,

$$\mathbf{C}^{\mathrm{CoDE}} = \left(C^{\mathrm{priv}}, C^{\mathrm{rate}}, C^{\mathrm{soc}}\right). \tag{1}$$

Each perspective assigns the delay cost to a bearer: C^{priv} to the developer or customer, C^{rate} to the utility and its ratepayers, and C^{soc} to any external cost not internalized by the other two. The three are not added directly, because C^{priv} contains transfers to counterparties outside the model, the split between

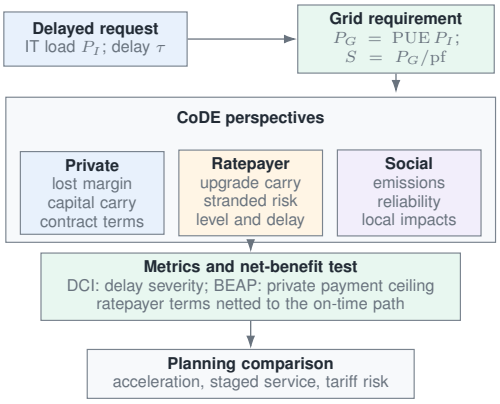


Fig. 1. CoDE framework for delayed grid energization.

C^{priv} and C^{rate} is itself a tariff choice, and C^{soc} is incomplete. Section II-E states the test under which they may be combined.

B. Private Perspective

The private perspective collects lost operating margin, carrying cost on committed capital, contract penalties, and mitigation premiums,

$$C^{\mathrm{priv}}(P_I, \tau) = \sum_{t=1}^{\tau} d_t\, P_I \Big[q(1-s) V \mu + \pi^{\mathrm{contract}} + \pi^{\mathrm{mit}} + K\, c\, \alpha_t\, i/12 \Big], \tag{2}$$

where $d_t = (1+r)^{-t/12}$ is the monthly discount factor at annual rate r; q the probability the project is realized; s the fraction of gross margin recovered through the developer's own arrangements during the delay, with any premium for them carried in π^{mit}; V gross value in USD per MW-IT-month; μ operating margin; K committed capital cost per MW-IT; i the annual carrying rate; c the committed capital fraction; and α_t the capital deployment profile with peak level a. Margin is weighted by q because it accrues only if the project proceeds. The carrying term takes no q weight as committed capital is financed either way, and abandonment is assumed to resolve after the delay window. Hence, carrying cost accrues over the full τ with profile $c\alpha_t$. The contract and mitigation terms are unweighted because penalties are triggered by the delay itself and mitigation premiums are paid while it lasts.

C. Ratepayer Perspective

The ratepayer perspective captures expected utility carrying cost for network upgrades and exposure to stranded assets,

$$C^{\mathrm{rate}}(\tau) = \sum_{t=1}^{\tau} d_t\, P_G\, U\, i_u\, \rho\, \phi_u/12 + d_\tau\, P_G\, U\, \rho\, \phi_u\, \phi_s, \tag{3}$$

where U is the utility upgrade cost per grid MW, i_u the annual utility carrying rate, ρ the residual ratepayer share after tariff protections, ϕ_u the probability an upgrade is financed during

the delay, and ϕ_s the stranding probability conditional on financing. A single ρ covers both terms; splitting it would let minimum bills and exit fees, which protect the carrying and stranding terms respectively, vary independently.

Unlike C^{priv}, which vanishes at $\tau = 0$, C^{rate} does not: the carrying sum is empty but the stranding term is not, so part of the level is exposure a financed request carries whatever its timing. The delay-attributable ratepayer cost is therefore the differential $\Delta C^{\mathrm{rate}}(\tau) = C^{\mathrm{rate}}(\tau) - C^{\mathrm{rate}}(0)$, and it is this quantity, not the level, that is comparable with C^{priv}.

D. Social Perspective

The social perspective is pathway-specific: it prices the external effects of whatever substitute fills the delay, such as bridge power, relocation, curtailment, or staged service. For a pathway h used with probability p_h, a generic representation is

$$C^{\mathrm{soc}}(h) = p_h \sum_{t=1}^{\tau} d_t \left(C_t^{\mathrm{em}}(h) + C_t^{\mathrm{rel}}(h) + C_t^{\mathrm{loc}}(h) \right), \quad (4)$$

where C_t^{em}, C_t^{rel}, and C_t^{loc} are monetized emissions, reliability, and local impact terms in month t. For an emissions-only pathway the first term becomes

$$C^{\mathrm{soc,em}}(h) = p_h \sum_{t=1}^{\tau} d_t \, E_t(h) \left[\epsilon_h(t) - \epsilon_0(t) \right] v_t, \quad (5)$$

where E_t is electricity served, $\epsilon_h - \epsilon_0$ is the pathway's incremental direct emission factor relative to the on-time counterfactual, and v_t is the social cost of CO_2. The counterfactual ϵ_0 should be the marginal grid intensity; the average factor used below is a public-data proxy for it, and $p_h = q$ holds only when every realized project uses pathway h.

E. Metrics and the Net-Benefit Test

Delay cost intensity expresses CoDE as cost per unit of delayed capacity per month, in USD per MW-IT-month for the private perspective and USD per grid-MW-month for the ratepayer perspective,

$$\mathrm{DCI}^{\mathrm{priv}} = \frac{C^{\mathrm{priv}}}{P_I \tau}, \quad \mathrm{DCI}^{\mathrm{rate}} = \frac{\Delta C^{\mathrm{rate}}}{P_G \tau}. \quad (6)$$

If a utility upgrade, staged energization agreement, flexible interconnection, relocation, or temporary supply arrangement m shortens the delay from τ_0 to $\tau_1(m)$, the avoided costs are

$$\Delta C_m^{\mathrm{priv}} = C^{\mathrm{priv}}(P_I, \tau_0) - C^{\mathrm{priv}}(P_I, \tau_1(m)), \quad (7)$$

with $\Delta C_m^{\mathrm{rate}}$ defined the same way from C^{rate}, and the added pathway externality $\Delta C_m^{\mathrm{soc}} = C^{\mathrm{soc}}(m) - C^{\mathrm{soc}}(\text{on time})$, normally positive. The $\tau = 0$ stock is common to both terms of $\Delta C_m^{\mathrm{rate}}$ and cancels, so no separate netting is required. The break-even acceleration payment is the scalar $\Delta C_m^{\mathrm{priv}}$, the most a developer can justify from avoided delay cost alone, and $[\, C_m^{\mathrm{add}},\ \mathrm{BEAP} \,]$ is the bargaining range against the implementation cost C_m^{add} borne by others; the option is privately worth doing when $\mathrm{BEAP} \geq C_m^{\mathrm{add}}$, under risk neutrality and ignoring the option value of waiting.

Option m is potentially welfare-improving when

$$\Delta C_m^{\mathrm{priv}} + \Delta C_m^{\mathrm{rate}} - C_m^{\mathrm{add}} - \Delta C_m^{\mathrm{soc}} > 0, \quad (8)$$

and distributionally acceptable when the developer's payment covers C_m^{add} plus compensation for $\Delta C_m^{\mathrm{soc}}$. Part of $\Delta C_m^{\mathrm{priv}}$ may be scarcity rent reallocated rather than created, so reporting it gross makes the left side of (8) an upper bound on the welfare gain. BEAP alone answers neither question; it is one input to both.

F. Staged Service

For staged energization, let a fraction f of IT capacity receive service after $\tau_e < \tau_0$ months while the rest waits until τ_0, and let $g(f)$ be the revenue-equivalent service fraction, which scales both the margin and the carrying term, with $g(f) = f$ in the benchmark. The private staged cost is

$$C_{\mathrm{stage}}^{\mathrm{priv}} = C^{\mathrm{priv}}(g(f)P_I, \tau_e) + C^{\mathrm{priv}}((1 - g(f))P_I, \tau_0), \quad (9)$$

and $C^{\mathrm{priv}}(P_I, \tau_0) - C_{\mathrm{stage}}^{\mathrm{priv}}$ is the private value of partial service.

III. Case Study

All benchmark values below are illustrative: ranges are assumed, cases are unweighted, no joint distribution is estimated, and results are sensitivity. Costs are in millions of USD (MUSD) and thousands of USD (kUSD).

A. Case Data and Benchmark Design

1) Benchmark Inputs: The benchmark evaluates 100, 300, and 500 MW-IT data centers delayed by 6, 12, 24, 36, and 60 months, with gross value anchored to CBRE's 196.25 USD/kW-month wholesale colocation value and 150 and 250 USD/kW-month sensitivities. Committed capital cases use JLL's shell-and-core value of 10.7 MUSD/MW-IT, which covers building and electrical works but not IT equipment, 35.7 MUSD/MW-IT with AI fit-out added, and an assumed midpoint of 20 MUSD/MW-IT. Table II summarizes the assumed parameters.

Capital deployment follows a static, linear, front-loaded, or back-loaded profile. Let $x_t = \min\{t/\tau, 1\}$ denote build progress, assumed to advance over the delay window, and let a be the peak level; the four profiles set α_t to a, ax_t, $a\min\{1, 2x_t\}$, and ax_t^2, respectively. The ratepayer benchmark assumes milestone-based financing: the utility commits to the upgrade with probability $\phi_u = 0.9$, set independently of q. Non-realization enters through the stranding probability conditional on financing, $\phi_s = (1 - q) + q\, p_a$, with p_a the probability a realized project's upgrade is stranded anyway. Contract and mitigation terms are zero.

2) Scenario Design: The base case is a 300 MW-IT project delayed 36 months, giving $P_G = 375$ grid MW and $S = 395$ MVA at PUE 1.25 and pf 0.95, with 196.25 USD/kW-month gross value, $\mu = 35\%$, $s = 25\%$, $q = 0.75$, $c = 100\%$, a static profile with $a = 50\%$, 8% discount and carrying rates, and $K = 10.7$ MUSD/MW-IT. Base ratepayer parameters are

TABLE II
KEY BENCHMARK PARAMETERS

Input	Values used	Model role
Capacity and delay	100, 300, 500 MW-IT; 6, 12, 24, 36, 60 mo.	Scenario scale
Gross value V	150, 196.25, 250 kUSD/MW-IT-mo.	Lost margin base
Margin and discounting	$\mu = 20, 35, 50\%$; $r = 6, 8, 10, 12\%$	Present value, private cost
Substitution and realization	$s = 0, 25, 50\%$; $q = 50, 75, 100\%$	Unserved realized load
Committed capital	$K =$ 10.7, 20, 35.7 MUSD/MW-IT; $c = 50, 75, 100\%$; $a = 50\%$ base, 25–75% sampled; four profiles; $i = 8\%$	Carrying cost
Grid conversion	PUE $= 1.25$ base, 1.15–1.35 sampled; pf $= 0.95$	MW-IT to MW and MVA
Tariff risk	$U =$ 1–2 MUSD per grid MW; $\rho =$ 0–100%; $\phi_u = 0.9$ base, 0.75–1.0 sampled; $p_a =$ 5–20%; $i_u = 8\%$	Ratepayer stress
Staged service	First service at 6, 12, 24 mo.; $f = 25, 50, 75, 100\%$	Partial service BEAP
Bridge pathway	85% utilization; 0.96 and 0.81 lb-CO_2/kWh; $v = 140, 230, 380$ USD/t-CO_2	Social externality
Excluded terms	$\pi^{\text{contract}} = \pi^{\text{mit}} = 0$	Require project data

$U = 1$ MUSD per grid MW, $\rho = 50\%$, $i_u = 8\%$, $p_a = 10\%$, and $\phi_u = 0.9$, giving $\phi_s = 0.325$.

The ten-dimensional full-factorial set varies capacity, delay, V, μ, r, s, q, K, c, and the deployment profile, giving 174,960 cases; realization probability and delay vary independently, so the percentiles of the set summarize the assumed scenario lattice, not a distribution over projects. A Latin hypercube sampling (LHS) screen of 1000 draws at $P_I = 300$ MW-IT and $\tau = 36$ months covers the continuous ranges in Table II and reports Spearman rank correlations.

A stress test replaces the base-case q with $q_\beta(\tau) = \min\{1, 0.75\exp[-\beta(\tau - 36)/12]\}$, where $\beta \in \{0, 0.05, 0.10\}$ yr^{-1} sets how fast realization probability declines per additional year of delay and all three curves pass through $q_\beta(36) = 0.75$. The staged extension evaluates first service at 6, 12, and 24 months for 25–100% of capacity, and a tariff stress test varies the residual ratepayer share, upgrade cost, and stranding probability.

B. Simulation Results

1) Magnitude and Decomposition: By construction, private DCI is independent of capacity; it declines with delay only through discounting, from 72.7 kUSD/MW-IT-month at 6 months to 66.1 at 36 and 61.5 at 60. In the 300 MW-IT case with a 36-month delay, lost margin is 371 MUSD and committed-capital carrying cost is 343 MUSD; the two are of comparable size. Of the 80 MUSD ratepayer level, 25 MUSD is delay-attributable, or 1.8 kUSD per grid-MW-month, and 55 MUSD is the stranding exposure a request carries whatever its timing (Fig. 2).

2) Sensitivity and Dependence: Capital-side inputs dominate private CoDE. The largest rank correlations are with the capital deployment profile dummies, which enter with opposite signs for the static and back-loaded profiles; margin, committed capital cost, and realization probability follow (Fig. 3). Ratepayer exposure is dominated by the residual ratepayer share, with realization probability next, entering negatively through the stranding term, then upgrade cost, PUE, and upgrade financing probability. The screen is run on levels, so the ratepayer bars include the stranding stock.

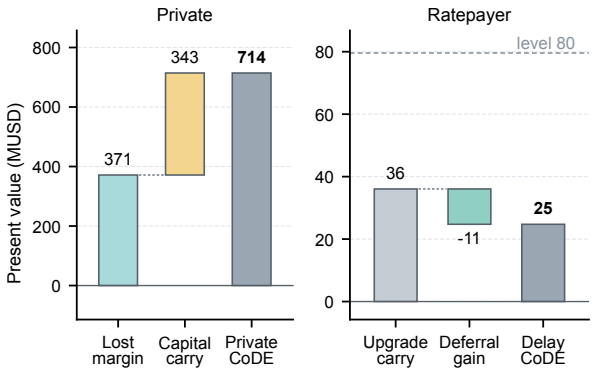


Fig. 2. Base-case decomposition for the 300 MW-IT project delayed 36 months. Each panel is a waterfall within one perspective, on independent scales, because CoDE does not aggregate across perspectives. The ratepayer panel is the delay differential; the dashed line marks the level, which includes the $\tau = 0$ stock.

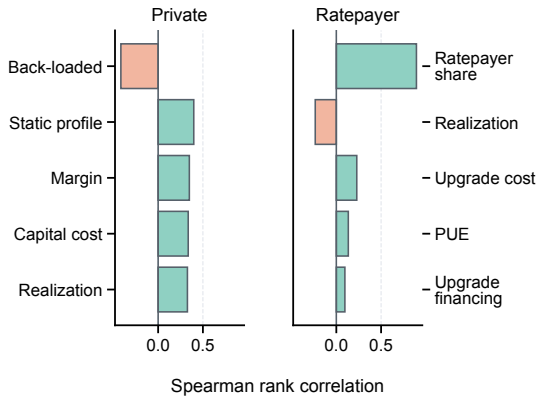


Fig. 3. Latin hypercube screening, 1000 samples at $P_I = 300$ MW-IT and $\tau = 36$ months. Bars are Spearman rank correlations, deployment profiles coded as dummies; the five largest by magnitude are shown, ordered by magnitude.

Across the 174,960 unweighted cases, median private CoDE rises from 113 MUSD at 6 months to 868 MUSD at 60 months, with wide within-delay dispersion.

Table III reports the stress in MUSD; all cases hold $q_\beta(36) = 0.75$, hence $C^{\text{priv}}(36) = 714.3$ MUSD. The delay column nets each level against an on-time path evaluated at $q_\beta(0) = 0.75$, 0.87, and 1, the cap binding at $\beta = 0.10$, so on-time service is credited with preventing delay-induced attrition as well; holding q at 0.75 instead isolates the timing effect and

TABLE III
DELAY–REALIZATION DEPENDENCE STRESS AT 60 MONTHS FOR THE 300 MW-IT BASE CASE

			Ratepayer	
β (yr^{-1})	$q_\beta(60)$	Private†	Level	Delay
0	0.750	1107	93.2	38.3
0.05	0.679	1052	100.6	64.1
0.10	0.614	1002	107.2	90.4

†Private CoDE vanishes at $\tau = 0$, so its level is its delay differential.



Fig. 4. Gross avoided private CoDE, before implementation cost and risk, from staged energization of the 300 MW-IT base case with a 36-month delay. Labels in MUSD.

gives 38 to 52 MUSD. Private CoDE decomposes as

$$C^{\mathrm{priv}}(\tau,\beta) = q_\beta(\tau)A_\tau + B_\tau,$$
$$\eta_{C,q} = \frac{\partial \ln C^{\mathrm{priv}}}{\partial \ln q} = \frac{q_\beta A_\tau}{q_\beta A_\tau + B_\tau}, \tag{10}$$

where $A_\tau = \sum_t d_t P_I (1-s) V \mu$ collects the terms weighted by realization probability and $B_\tau = \sum_t d_t P_I K c \alpha_t i/12$ is the carrying cost. At 60 months and $\beta = 0$, $q_\beta A_\tau = 575.4$ MUSD and $B_\tau = 531.2$ MUSD, giving $\eta_{C,q} = 0.520$. The informative quantity is the elasticity: carrying cost bounds how far realization risk can move private CoDE, so the 18.1% reduction in q at $\beta = 0.10$ moves it by only 9.4%. Ratepayer exposure moves the other way, because milestone financing leaves the carrying term unchanged while a lower q raises the stranding term: the level rises 15.1% and the delay differential rises from 38 to 90 MUSD. Under a realization-linked specification ($\phi_u = q$, $\phi_s = p_a$) the level would instead fall by the full 18.1%. Stranding is booked as one write-off at τ; booking in-window failures at their failure dates would raise the $\beta = 0.10$ stranding term by about one fifth and its level by about a tenth, so trailing digits are not meaningful.

3) Staged Service and Tariff Tests: When full early service is infeasible, partial energization still captures much of the acceleration value: energizing 50% of capacity after 12 months avoids 229 MUSD of private CoDE, 75% avoids 343 MUSD, and full service after 12 months avoids 458 MUSD (Fig. 4).

These are gross private thresholds and exclude the implementation cost C_m^{add} and risk. Avoided ratepayer exposure for the same options is 1.9–20.2 MUSD, already net of the stranding stock and an order of magnitude below the private figures, so a high BEAP is no evidence of comparable public benefit; passing (8) depends on C_m^{add} and the pathway externality. Earlier service also brings the write-off forward: at full service after 6 months a 29.5 MUSD carrying saving nets to 20.2 MUSD.

With base private CoDE fixed at 714 MUSD, the ratepayer level is zero when upgrades are developer-funded, 80 MUSD in the base case ($\rho = 50\%$, $U = 1$ MUSD/MW, $p_a = 10\%$), and 359 MUSD at the range maxima ($\rho = 100\%$, $U = 2$ MUSD/MW, $p_a = 20\%$), with delay-attributable parts of zero, 25, and 89 MUSD. The level is what tariff design allocates; the differential is what acceleration avoids.

4) Bridge Power as a Priced Pathway: A natural-gas bridge is simultaneously a mitigation option and a source of externality, so both sides are evaluated against the same on-time counterfactual. Serving the base 375 MW grid load for the first 12 months of the 36-month delay consumes 2.792 TWh at 85% utilization and avoids 257 MUSD of private CoDE. Taking the 2023 U.S. utility-scale average direct emission factors of 0.96 lb-CO_2/kWh (0.44 kg/kWh) for natural gas and 0.81 lb-CO_2/kWh for all generation as marginal proxies, the bridge emits 0.190 Mt-CO_2 more than on-time grid service [23]. Valuing those emissions at the EPA 2030 social costs of CO_2 of 140, 230, and 380 USD per tonne (2020 dollars; 2.5%, 2.0%, and 1.5% near-term Ramsey rates), with every realized project using the bridge ($p_h = q = 0.75$), gives ex ante damages of 19.9, 32.8, and 54.1 MUSD [24], or 25–68 MUSD in current dollars at a price factor of about 1.25. Avoided private cost therefore exceeds the climate term by a factor of about 4 to 10. Discounting within the 12-month window moves the damage term by about 4%, so (5) uses $d_t = 1$.

The bridge avoids no ratepayer exposure, since the upgrade schedule is unaffected, so it passes (8) whenever its power-cost premium C_m^{add} is below the headroom of 232 MUSD at the low social cost of CO_2 to 189 MUSD at the high one, or 68–83 USD/MWh of bridged energy. The direct-CO_2 term changes sign if the marginal grid emission factor exceeds the bridge factor of 0.96 lb-CO_2/kWh; and if the bridge raises realization probability it also lowers the ratepayer stranding term, a credit the benchmark omits.

C. Discussion and Takeaways

1) Scope of the Benchmark: The bridge case monetizes direct CO_2 differences only, excluding upstream methane, local pollution, water, and reliability effects. Avoided private cost credits the full carrying saving in every state, consistent with abandonment resolving after the delay window; crediting it only in realization states would lower every avoided-cost figure by about one eighth, to 226 and 403 MUSD for the bridge and for full service after 12 months. Because the benchmark assumes milestone-based financing, ratepayer exposure rises as realization probability falls; if the utility instead commits only once the project goes forward ($\phi_u = q$, $\phi_s = p_a$), the sign reverses. Which case applies follows from whether irreversible network commitment precedes or follows project resolution, so

the sign diagnoses commitment timing; intermediate structures give intermediate signs.

2) Planning Takeaways:

- *For developers:* lost margin and carrying cost are comparable, 371 and 343 MUSD, so acceleration value does not rest on the revenue term alone.
- *For utilities:* staged-service BEAP gives a defensible ceiling against which acceleration cost and risk can be compared.
- *For regulators:* tariff design moves the ratepayer level between zero and 359 MUSD, of which zero to 89 MUSD is delay-attributable, without touching private delay cost.

3) Limitations and Extensions: CoDE takes the timing and quantity of usable service as given and prices the consequences; it does not model why service is late. The MW/MVA conversion gives request scale only; bus-level feasibility, thermal and voltage limits, contingencies, and lead times require a network study whose results become CoDE inputs. The upgrade is charged at full cost from month one while developer capital follows a deployment profile; giving it its own profile would lower the ratepayer carrying term. C^{priv} counts within-window costs only, so continuation value lost to attrition is excluded and the private column is conservative for $\beta > 0$. Applications should replace the assumed ranges with service-request, project, tariff, and pathway data.

IV. Conclusion

CoDE prices grid service that arrives later than planned, keeps the developer, ratepayer, and social accounts separate, and states the test under which they may be combined. Two qualitative results follow from this structure; their magnitudes depend on the assumed ranges. First, committed-capital carrying cost is independent of realization probability and supplies about half of private CoDE at long delays, so an 18.1% fall in that probability lowers private CoDE by only 9.4%, leaving it above 1.0 billion USD at 60 months, while under milestone financing the same fall raises delay-attributable ratepayer exposure from 38 to 90 MUSD. Second, the acceleration options studied barely move the third-party accounts, at most 20.2 MUSD of avoided ratepayer exposure and 25–68 MUSD of bridge climate damage against 229–458 MUSD of private value, even though the ratepayer level itself spans zero to 359 MUSD across tariff designs. What remains open for these options is funding and compensation, and applications require project-, tariff-, and pathway-specific data.

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